

# Introduction to Statistics

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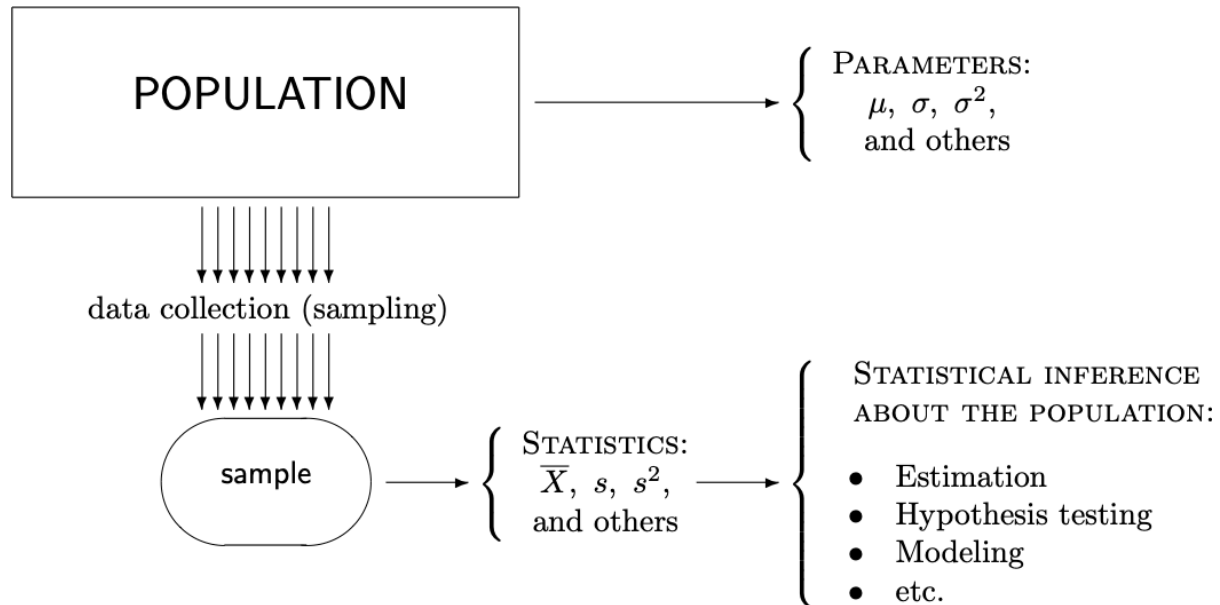
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# Probability Theory and Data

Probability theory underpins randomness.

In real-life, we need data to understand probability distributions.



# Sample and Statistic

Let  $p$  be a pmf/pdf. We call  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p$  a **sample** of  $p$ .

- $p$  is often called **population** when samples are involved.
- iid = **independent and identically distributed**.
- $n$  is called the **sample size**.
- A function of  $(X_1, \dots, X_n)$  is called a **statistic**.

We can use the values of some specific statistics to describe  $p$ .

## Definition

Let  $F$  be the cumulative distribution function (cdf) of a population (pmf or pdf)  $p$ . The **median** of  $p$  is a number  $M$  such that

$$P(x < M) \leq \frac{1}{2} \quad \text{and} \quad P(x > M) \leq \frac{1}{2}$$

## Remarks

- If  $p$  is a pdf, then  $M$  is a unique value such that

$$F(M) = \int_{-\infty}^M p(x) dx = \frac{1}{2}.$$

- If  $p$  is a pmf, then  $M$  may not be unique.

## Example

Find the median of  $\text{Exponential}(\lambda)$ .

## Solution

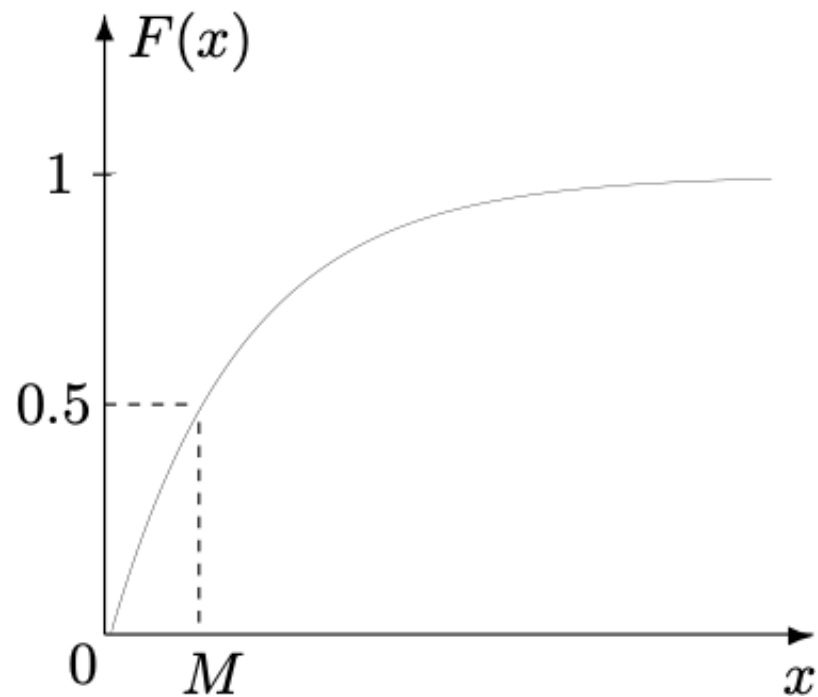
Recall the pdf of  $\text{Exponential}(\lambda)$  is  $p(x) = \lambda e^{-\lambda x}$  for  $x > 0$  and 0 elsewhere.

Hence the cdf is

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & x > 0, \\ 0 & x \leq 0. \end{cases}$$

Solving  $F(M) = 1 - e^{-\lambda M} = \frac{1}{2}$  yields the median  $M = \frac{\ln 2}{\lambda}$ .

The cdf  $F$  and the median  $M$  in the Exponential( $\lambda$ ) example

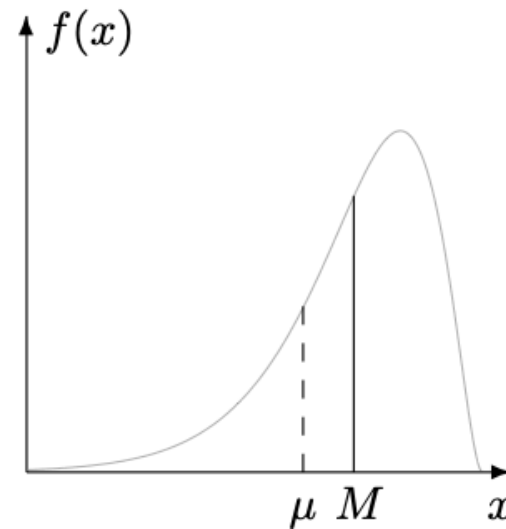
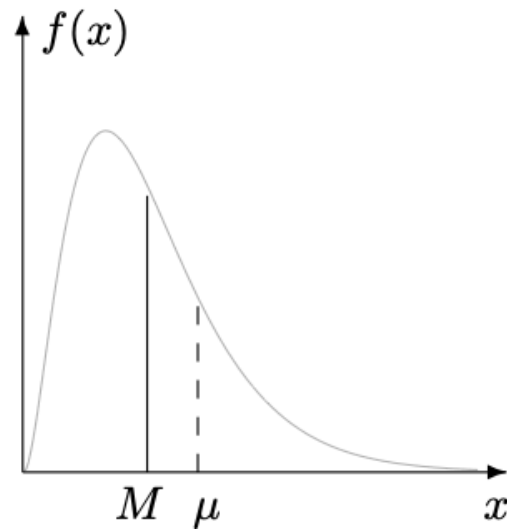
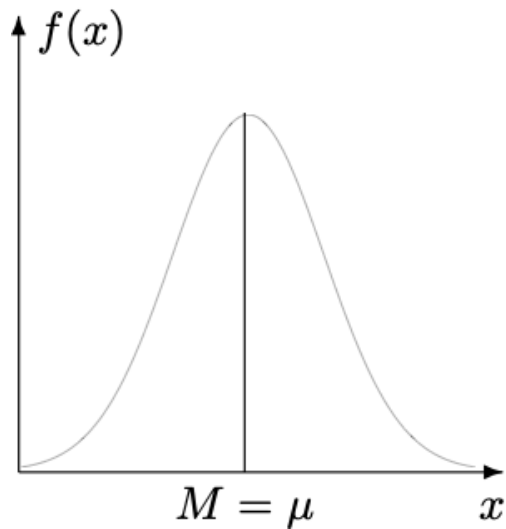


| Compare median $M$ and mean $\mu$ | $M = \mu$ | $M < \mu$    | $M > \mu$   |
|-----------------------------------|-----------|--------------|-------------|
| Skewness of the population $p$    | symmetric | right-skewed | left-skewed |

(a) *symmetric*

(b) *right-skewed*

(c) *left-skewed*



**Example:** Exponential( $\lambda$ ) has  $M = \frac{\ln 2}{\lambda} \approx \frac{0.69}{\lambda}$  and  $\mu = \frac{1}{\lambda}$ , so it is right-skewed.

## Example

Find the median of Binomial(5,0.5).

## Solution

Recall the pmf is  $p(x) = \binom{5}{x} (0.5)^x (1 - 0.5)^{5-x}$  for  $x = 0, 1, \dots, 5$ . Then

$$F(x) = \sum_{k \leq x} p(k)$$

By calculations, we see the median  $M$  can be any value in the interval  $(2, 3)$ .



## Example

Find the median of Binomial(5,0.4).

## Solution

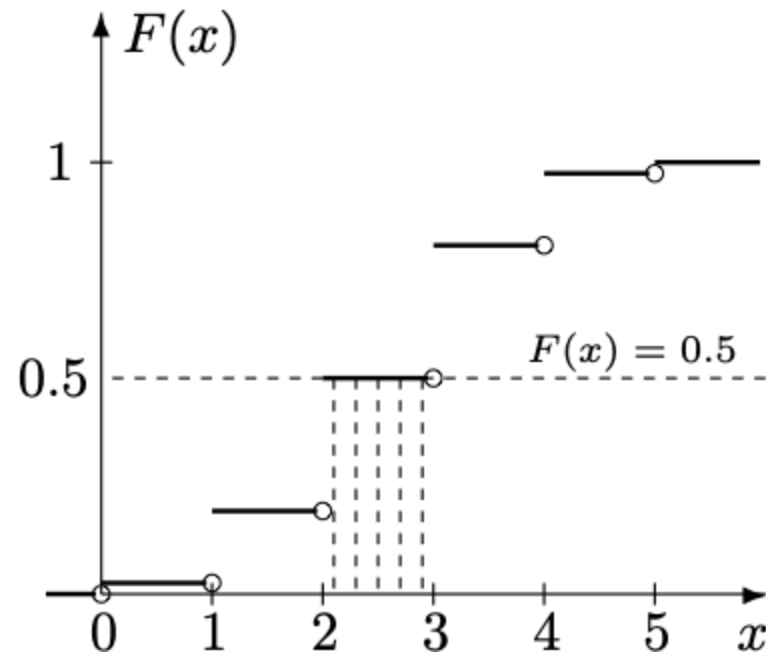
Recall the pmf is  $p(x) = \binom{5}{x} (0.4)^x (1 - 0.4)^{5-x}$  for  $x = 0, 1, \dots, 5$ . Then

$$F(x) = \sum_{k \leq x} p(k)$$

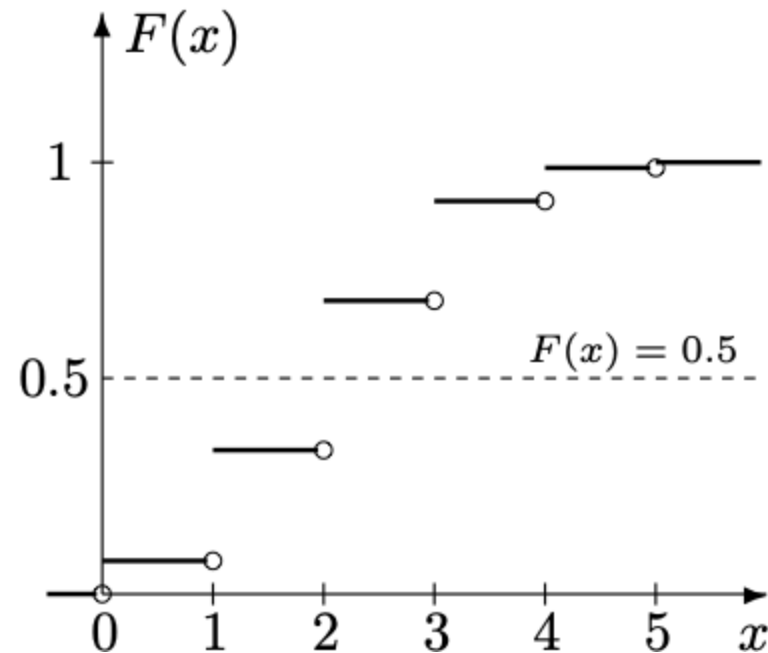
By calculations, we see only  $M = 2$  is the median.

## The cdf's $F$ of the two Binomial distributions above

(a) Binomial ( $n=5, p=0.5$ )  
many roots



(b) Binomial ( $n=5, p=0.4$ )  
no roots



## Definition

Let  $q \in (0, 1)$ , the  $q$ -**quantile** of a population  $p$  is a number  $x_q$  such that

$$P(X < x_q) \leq q \quad \text{and} \quad P(X > x_q) \leq 1 - q.$$

The following three  $q$ 's are mostly used:

|          |       |       |       |
|----------|-------|-------|-------|
| $q$      | 0.25  | 0.5   | 0.75  |
| Notation | $Q_1$ | $Q_2$ | $Q_3$ |

## Remarks

- $Q_2$  is just the median  $M$ .
- If  $p$  is a pdf, then  $Q_1$  is the value such that  $F(Q_1) = \int_{-\infty}^{Q_1} p(x)dx = 0.25$ , etc.

## Example

Find  $Q_1$  and  $Q_3$  of Exponential( $\lambda$ ).

## Solution

Solving  $F(Q_1) = 1 - e^{-\lambda Q_1} = 0.25$  yields  $Q_1 = \frac{\ln(4/3)}{\lambda}$ .

Solving  $F(Q_3) = 1 - e^{-\lambda Q_3} = 0.75$  yields  $Q_3 = \frac{\ln 4}{\lambda}$ .

## Definition

Let  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p$  be a sample and sorted in ascending order, then its **sample median**, denoted by  $\widehat{M}$ , is defined by

- $\widehat{M} = X_{\frac{n+1}{2}}$  if  $n$  is odd.
- $\widehat{M} = \frac{1}{2} (X_{\frac{n}{2}} + X_{\frac{n}{2}+1})$  if  $n$  is even.

## Example

Given a sample 3, 5, 2, 9, 2. Find its sample median.

## Solution

Sort the sample in ascending order:

2, 2, 3, 5, 9

Since the sample size  $n = 5$  is odd, the sample median is

$$\widehat{M} = X_{\frac{n+1}{2}} = X_3 = 3.$$

## Example

Given a sample 3, 5, 2, 9, 2, 9. Find its sample median.

## Solution

Sort the sample in ascending order:

2, 2, 3, 5, 9, 9

Since the sample size  $n = 6$  is even, the sample median is

$$\widehat{M} = \frac{1}{2}(X_{\frac{n}{2}} + X_{\frac{n}{2}+1}) = \frac{1}{2}(X_3 + X_4) = \frac{1}{2}(3 + 5) = 4.$$

## Definition

Let  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p$  be a sample and sorted in ascending order and  $q \in (0, 1)$ , then its **sample  $q$ -quantile**, denoted by  $\hat{X}_q$ , is the value in this sample such that

- $\hat{X}_q$  is larger than at most  $q$  percent of the sample.
- $\hat{X}_q$  is smaller than at most  $(1 - q)$  percent of the sample.

Again, the three  $q$ 's are mostly used with shorthand notations (e.g.,  $\hat{Q}_1 = \hat{X}_{0.25}$ ):

| $q$      | 0.25        | 0.5         | 0.75        |
|----------|-------------|-------------|-------------|
| Notation | $\hat{Q}_1$ | $\hat{Q}_2$ | $\hat{Q}_3$ |

Note  $\hat{Q}_2 = \hat{M}$ .



## Example

Find  $\hat{Q}_1$ ,  $\hat{Q}_2$ , and  $\hat{Q}_3$  of the following sample (already sorted):

|    |    |           |    |           |           |    |           |    |     |
|----|----|-----------|----|-----------|-----------|----|-----------|----|-----|
| 9  | 15 | 19        | 22 | 24        | 25        | 30 | <u>34</u> | 35 | 35  |
| 36 | 36 | 37        | 38 | <u>42</u> | <u>43</u> | 46 | 48        | 54 | 55  |
| 56 | 56 | <u>59</u> | 62 | 69        | 70        | 82 | 82        | 89 | 139 |

## Solution

Sample size is  $n = 30$ . Hence  $0.25n = 7.5$  and  $0.75n = 22.5$ . Just need to round them up to 8 and 23, respectively, to get

$$\hat{Q}_1 = X_8 = 34 \quad \text{and} \quad \hat{Q}_3 = X_{23} = 59.$$

In addition,  $\hat{Q}_2 = \hat{M} = \frac{1}{2}(X_{15} + X_{16}) = 42.5$ .

## Outlier Detection

The **sample interquartile range**  $\widehat{\text{IQR}}$  is defined by

$$\widehat{\text{IQR}} = \hat{Q}_3 - \hat{Q}_1$$

The **1.5(IQR) rule** says any number outside of

$$[\hat{Q}_1 - 1.5(\widehat{\text{IQR}}), \hat{Q}_3 + 1.5(\widehat{\text{IQR}})]$$

is considered as an **outlier**.

## Example

Consider the previous sample with  $\hat{Q}_1 = 34$  and  $\hat{Q}_3 = 59$ . So

$$\widehat{\text{IQR}} = \hat{Q}_3 - \hat{Q}_1 = 59 - 34 = 25.$$

The end points of the IQR interval are

$$\hat{Q}_1 - 1.5(\widehat{\text{IQR}}) = -3.5, \quad \hat{Q}_3 + 1.5(\widehat{\text{IQR}}) = 96.5.$$

Note that 139 in the sample is outside of  $[-3.5, 96]$  and hence considered as an outlier.

## Definition

Let  $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} p$  be a sample. Its **sample variance**, denoted by  $s^2$ , is defined by

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Its **sample standard deviation** is defined by  $s$ .

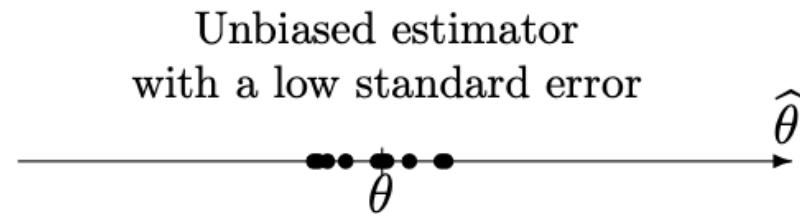
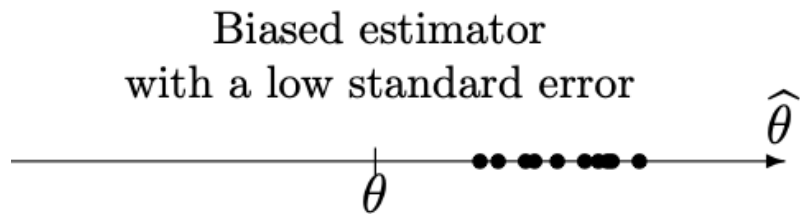
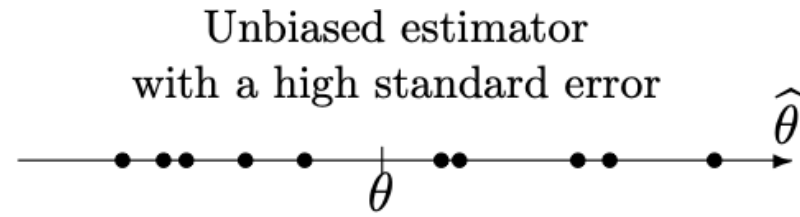
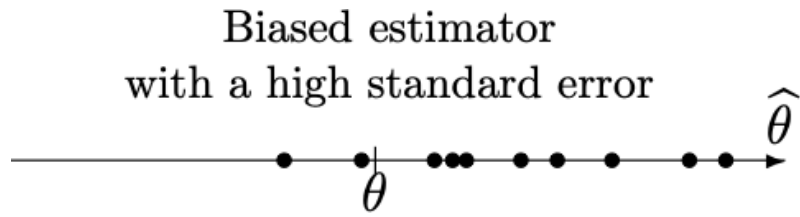
## Remarks

- $s^2$  is an unbiased estimator of variance, i.e.,  $\mathbb{E}(s^2) = \sigma^2$ .
- After simple algebra, we can show  $s^2 = \frac{1}{n-1} (\sum_{i=1}^n X_i^2 - n\bar{X}^2)$ .

Let  $\hat{\theta}$  be an estimator of  $\theta$ . Two quantities give a quick quality evaluation of  $\hat{\theta}$ :

- **Biasness:**  $\mathbb{E}(\hat{\theta}) - \theta$ .
- **Standard error:**  $\sigma(\hat{\theta})$ .

**Example:** 10 samples each giving an estimate (show as  $\bullet$ )  $\hat{\theta}$ :



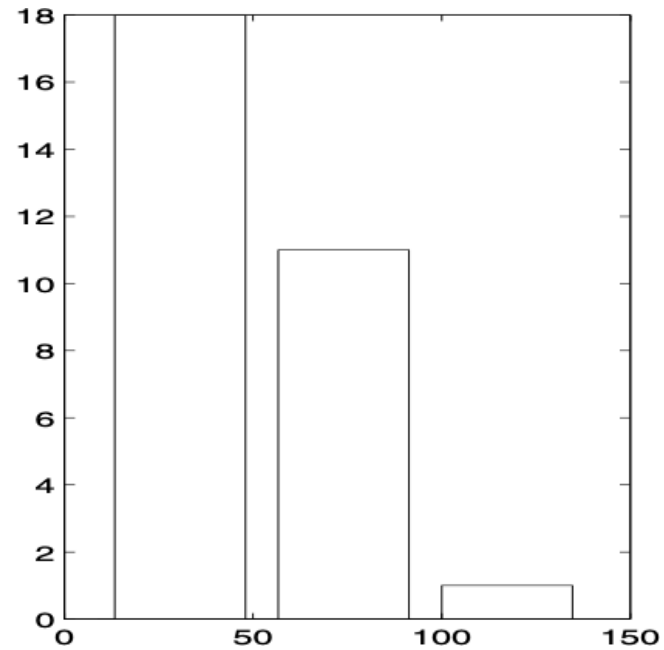
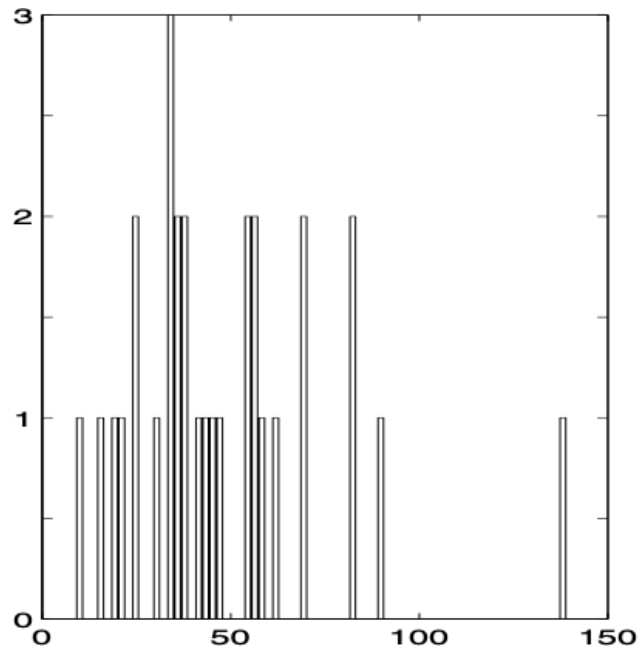
## Histogram

Histogram is a plot of  $p$  using a sample  $X_1, \dots, X_n$ :

1. Specify a proper bin size.
2. For each bin, draw a bar with height equal to the number of observed values following into the bin.
3. If a relative frequency histogram is needed, just scale all bar heights by  $\frac{1}{n}$ .

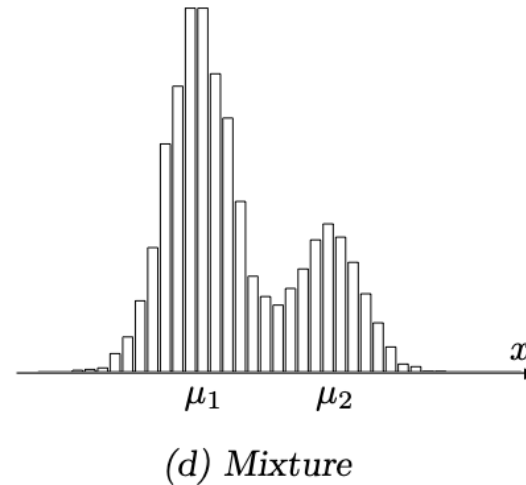
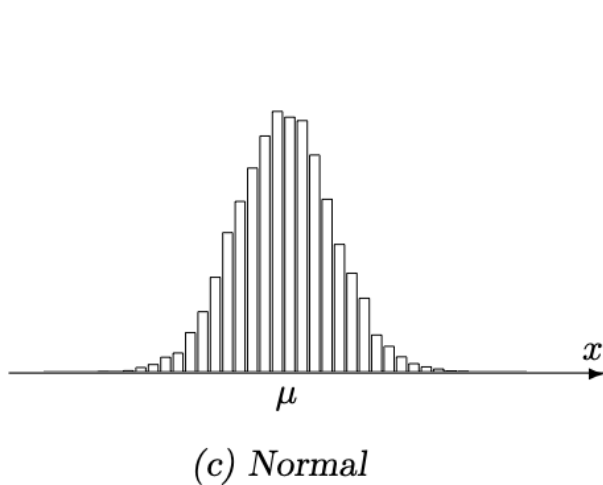
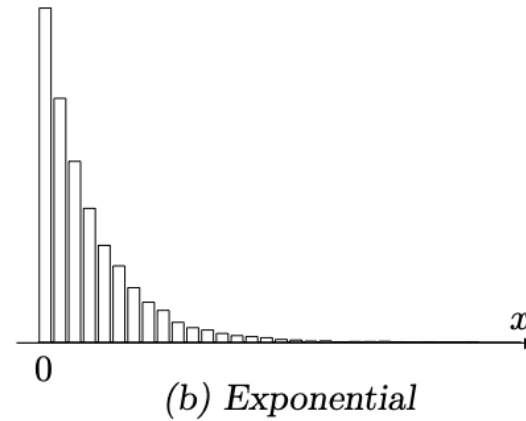
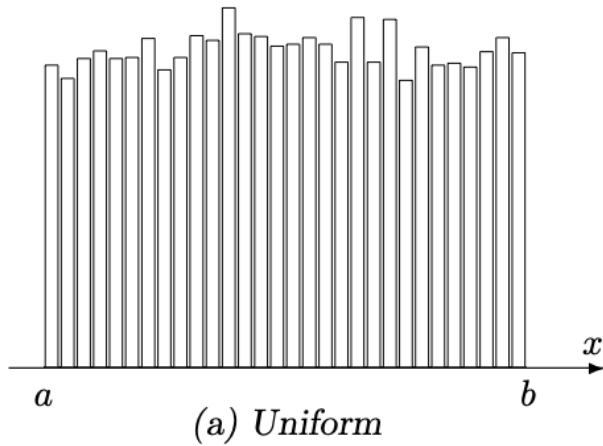
## Bin size is critical in drawing histograms

Histograms with bin size too small or too large:



## Histograms of some typical distributions:

Histogram of a sample can give some idea about the population  $p$ .



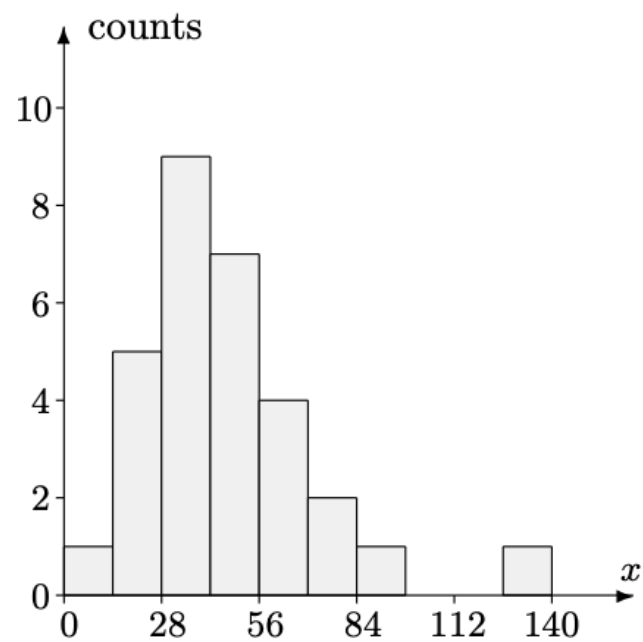


## Example

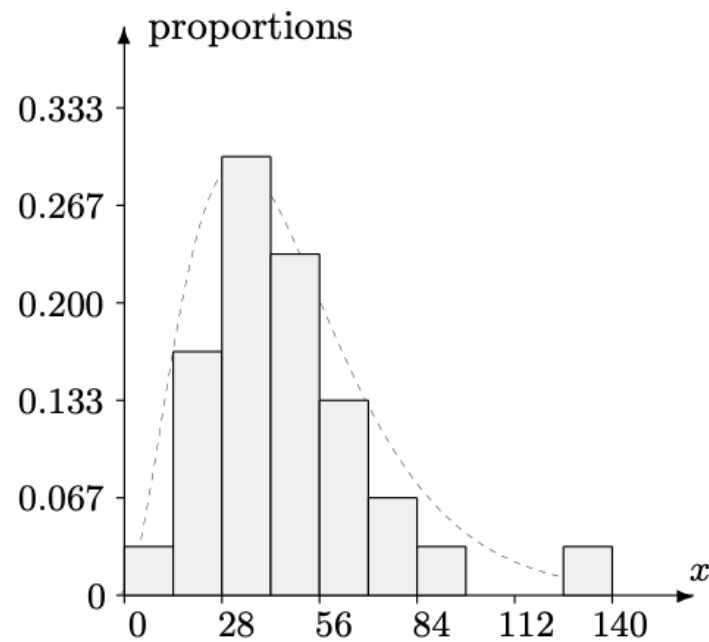
Consider the sample in the previous example:

|    |    |           |    |           |           |    |           |    |     |
|----|----|-----------|----|-----------|-----------|----|-----------|----|-----|
| 9  | 15 | 19        | 22 | 24        | 25        | 30 | <u>34</u> | 35 | 35  |
| 36 | 36 | 37        | 38 | <u>42</u> | <u>43</u> | 46 | 48        | 54 | 55  |
| 56 | 56 | <u>59</u> | 62 | 69        | 70        | 82 | 82        | 89 | 139 |

Choose bin size = 14, and draw the histograms of the sample:



(a) Frequency histogram



(b) Relative frequency histogram

## Stem-and-Leaf plot

Specify the stem and leaf units, and draw "histogram" with sample values. For example, set the two units as 10 and 1, write each value as (10 "stem" + 1 "leaf") form and put in the plot:

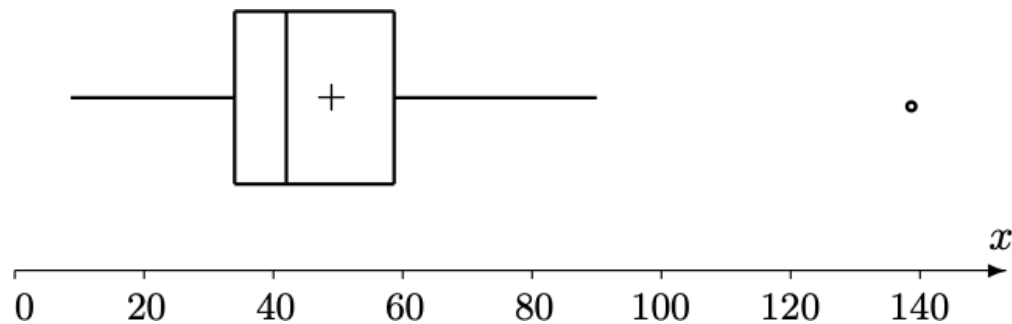
|               |    |   |   |   |   |   |   |   |   |
|---------------|----|---|---|---|---|---|---|---|---|
| LEAF UNIT = 1 | 0  | 9 |   |   |   |   |   |   |   |
|               | 1  | 5 | 9 |   |   |   |   |   |   |
|               | 2  | 2 | 4 | 5 |   |   |   |   |   |
|               | 3  | 0 | 4 | 5 | 5 | 6 | 6 | 7 | 8 |
|               | 4  | 2 | 3 | 6 | 8 |   |   |   |   |
|               | 5  | 4 | 5 | 6 | 6 | 9 |   |   |   |
|               | 6  | 2 | 9 |   |   |   |   |   |   |
|               | 7  | 0 |   |   |   |   |   |   |   |
|               | 8  | 2 | 2 | 9 |   |   |   |   |   |
|               | 9  |   |   |   |   |   |   |   |   |
|               | 10 |   |   |   |   |   |   |   |   |
|               | 11 |   |   |   |   |   |   |   |   |
|               | 12 |   |   |   |   |   |   |   |   |
|               | 13 | 9 |   |   |   |   |   |   |   |

## Boxplot

Boxplot is a plot of the so-called **five-point summary** of a sample  $X_1, \dots, X_n$ :

$$(\min, \hat{Q}_1, \hat{M}, \hat{Q}_2, \max)$$

- min (max) is the smallest (largest) value of  $X_1, \dots, X_n$ .
- Box left and right sides are  $\hat{Q}_1$  and  $\hat{Q}_3$ . Box middle line is  $\hat{M}$ .
- Sign "+" is  $\bar{X}$ . Horizontal line is IQR interval  $[\hat{Q}_1 - 1.5(\widehat{\text{IQR}}), \hat{Q}_2 + 1.5(\widehat{\text{IQR}})]$ .
- Values out of this interval (outliers) are plotted as "o".



## Remarks

- Sample mean, sample variance, and sample standard deviation are sensitive to outliers.
- $\hat{Q}_1, \hat{M}, \hat{Q}_2$  are much less sensitive to outliers.

## Parallel boxplots

Multiple samples can be plotted as parallel boxplot for comparison:

